SECTION 2: DATA COMPRESSION & RUN-LENGTH ENCODING [10 MARKS]

4. Lossy vs Lossless Comparison: (4 Marks)

Complete the comparison table by ticking [✓] or stating True/False:

Statement / Scenario	Lossy	Lossless
Decompressed file is 100% identical to the original uncompressed file.	**No**	**Yes [✓]**
Permanently removes background sounds that human ears cannot perceive.	**Yes [✓]**	**No**
Suitable for executable software program code and medical records.	**No**	**Yes [✓]**
Audio format MP3 and image format JPEG.	**Yes [✓]**	**No**

5. Run-Length Encoding (RLE) Problem Solving: (6 Marks)

a) A sequence of pixel colours is stored as: `W W W W W B B B B G G G G G G W W`

Write the RLE compressed string using (count, colour) pairs: (3 Marks)

Answer: `5W 4B 6G 2W` (or `05 W 04 B 06 G 02 W`). Original = 17 bytes, RLE = 8 bytes. [3 marks]

b) An RLE string is given as: `04 R 03 Y 05 B 02 G`

Decompress the string into its original sequence: (2 Marks)

Answer: `R R R R Y Y Y B B B B B G G` [2 marks]

c) Under what condition does RLE fail to reduce file size? (1 Mark)

Answer: When there are few or no consecutive repeating characters (e.g. `A B C D E F`), each character requires a count byte, doubling the size. [1 mark]

SECTION 3: ERROR DETECTION METHODS [10 MARKS]

6. Parity Block Matrix Error Location: (5 Marks)

A computer transmits data using **Even Parity**. The received parity block grid is shown below:

Byte	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Parity Bit
Byte 1	1	0	1	1	0	0	1	0
Byte 2	0	1	1	0	1	1	0	0
Byte 3	1	1	0	1	0	1	1	1
Byte 4	0	0	1	**0**	1	0	1	1
Parity Byte	0	0	1	**1**	0	0	1	0

a) Identify which byte contains the transmission error: **Byte 4 (has 3 ones with even parity bit 1 → total 4, wait: 0+0+1+0+1+0+1 = 3 ones, parity 1 → Odd parity error in Row 4!) [2 marks]**

b) Identify which bit column flipped: **Bit 4 (column 4 sum = 1+0+1+0+1 = 3 ones, parity 1 → parity error!)**